FlagForge

Interactive World Atlas & Vexillology Quiz Arena

A Project Report

Submitted in partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

By

Kaushal Rajput

Under the esteemed guidance of

_____

DEPARTMENT OF COMPUTER SCIENCE

_____ COLLEGE

(Affiliated to _____ University)

2026 - 2027

CERTIFICATE

This is to certify that

Kaushal Rajput

has satisfactorily completed the project report entitled

FlagForge - Interactive World Atlas & Vexillology Quiz Arena

in partial fulfilment of the requirements for the Bachelor of Science (Computer Science) during the academic year 2026 - 2027.

Date of Submission: _____

Project Guide	Co-ordinator	Examiner
Head of Department	Principal	College Seal

Declaration

I hereby declare that the project entitled **"FlagForge - Interactive World Atlas & Vexillology Quiz Arena"**, submitted in partial fulfilment for the award of Bachelor of Science in Computer Science during the academic year 2026 - 2027, represents my project work. Sources and third-party services used in the implementation are acknowledged in the references.

Signature of the Student: _____

Place: _____

Date: _____

Acknowledgement

I would like to express my sincere gratitude to everyone who contributed to the successful creation and development of the "FlagForge" project.

I extend my deepest gratitude to the faculty members and the project guide of the Department of Computer Science for their guidance and support during the development of this project. Their insights greatly enhanced its quality.

I also thank my friends and classmates for providing feedback on the quiz, explorer and comparison experiences. Their diverse perspectives helped me refine the functionality and the user experience of the application.

Building FlagForge helped me understand how a real-world application combines external information services, a responsive user interface, typed data validation, client-side state, accessibility and software testing. It was a rewarding and insightful experience.

Finally, I thank my family for their encouragement throughout the project.

Thank you all.

Kaushal Rajput

Abstract

FlagForge is a web-based interactive world atlas and geography quiz platform. It helps learners explore every country and territory in the world, practise recognising flags, capitals and borders through timed quizzes, take a shared daily challenge, and compare countries side by side using demographic data and charts.

Country data is aggregated from the REST Countries API, Wikipedia and Open-Meteo, validated with Zod schemas and normalized into one internal `Country` model. If the network is unavailable, an embedded dataset of 30 countries keeps the application usable. The application is a Progressive Web App: a service worker caches flags, API responses and visited pages so that previously loaded content works offline.

The application is built with Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS 4, TanStack Query and Zustand. All user data (progress, streaks, achievements, favorites and settings) is stored locally in the browser with versioned persistence, so no account is required and no personal data leaves the device. A pure, seeded quiz engine generates six question types at three difficulty levels and a deterministic "Daily Challenge" that is identical for every player on the same UTC date.

The project emphasizes keyboard accessibility (WCAG 2.2 AA target), graceful handling of API failures, a strict Content-Security-Policy, and automated verification with Vitest (58 unit and component tests) and Playwright with axe-core (end-to-end and accessibility specs).

Keywords: world atlas, geography quiz, flags, Next.js, React, PWA, offline-first, Zustand, TanStack Query, accessibility.

Aspect	Summary
Users	Students, geography enthusiasts and casual players (no account needed)
Platform	Responsive, installable Progressive Web App (desktop and mobile browsers)
Data sources	REST Countries v3.1, Wikipedia REST summary API, Open-Meteo, FlagCDN images
Storage	Browser localStorage (versioned Zustand stores), TanStack Query cache, service-worker caches
Verification	Vitest + Testing Library, Playwright, axe-core, TypeScript strict mode, ESLint

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1. Project Overview

FlagForge is a fast, offline-capable interactive world atlas, flag and geography quiz arena, and country comparison tool. All content is presented through a single responsive interface that works on phones as well as desktops.

Key Features

1. **World Atlas Explorer:** Browse every country and territory. Instant fuzzy search (Fuse.js) by name or capital, filters for region and subregion, population range, landlocked and UN-member status, grid or list view, and favorites.
2. **Quiz Arena:** Six quiz modes (flag to country, country to capital, capital to country, higher or lower, which country borders, map location) at three difficulty levels (easy, medium, hard) plus "all". Timed questions, speed bonuses, streak multipliers up to 3.0x and an end-of-game review.
3. **Daily Global Challenge:** A 10-question quiz generated from a seed derived from the UTC date, so every player receives identical questions each day. Results can be shared as an emoji grid.
4. **Compare:** Select up to three countries and compare population, area, borders, languages and currencies in a matrix and in Recharts bar and radar charts.
5. **Stats Dashboard:** Streaks, accuracy by region (radar chart), an activity heatmap calendar, weakest countries and unlockable achievements.
6. **Country Profiles:** A page per country with key facts, borders, a map card, live capital-city weather (Open-Meteo) and a Wikipedia summary.
7. **Productivity and Accessibility Extras:** Command palette (Ctrl/Cmd + K), keyboard answers (1-4), pause with Esc, light/dark/system theme, high-contrast and reduced-motion settings, installable PWA with offline fallback.

The scope of the system is geography learning and exploration. User accounts, multiplayer leaderboards and server-side storage are outside the scope of this project.

Feature	Implemented in	Route
Atlas explorer	src/components/features/explorer; use-explorer-filter.ts	/explore
Quiz arena	src/lib/quiz; src/components/features/quiz	/play
Daily challenge	src/lib/quiz/rng.ts; DailyModal	/daily
Compare	src/components/features/compare	/compare
Statistics	src/components/features/stats; useProgressStore	/stats
Country profile	src/components/features/country; src/lib/api	/country/[code]
Design reference	src/app/design/page.tsx	/design

2. Introduction & Motivation

Learning geography from textbooks and atlases is often passive. Learners may be able to look up a country but have little way to test themselves, track improvement, or compare nations quickly. Existing quiz sites often require registration, hide data behind advertising, or stop working without a connection.

FlagForge provides one interface for exploration, practice and comparison. It gathers country data from public APIs, offers six kinds of quiz, a shared daily challenge, a side-by-side comparison view and a personal statistics dashboard. The design emphasizes a clean, search-first interface, accessible controls and fast loading.

The system uses modern web technologies: Next.js 15 and React 19 with TypeScript for the application, Tailwind CSS 4 for styling, TanStack Query for server-state caching and Zustand for persisted client state. The same codebase renders fast pre-built pages and interactive client components.

The main motivation is practical. A learner may want a five-minute daily practice routine, may want to revise only a single region, or may want to know how two countries differ. Filters, difficulty levels and persistent progress support these needs, while streaks and achievements encourage regular practice.

Another motivation is low friction. No account is needed; progress is stored on the device. Because the app is installable and caches data, it remains useful on slow or unreliable connections.

Finally, building FlagForge is a realistic software-engineering exercise: it combines external APIs, schema validation, state management, offline caching, accessibility and automated testing in one project.

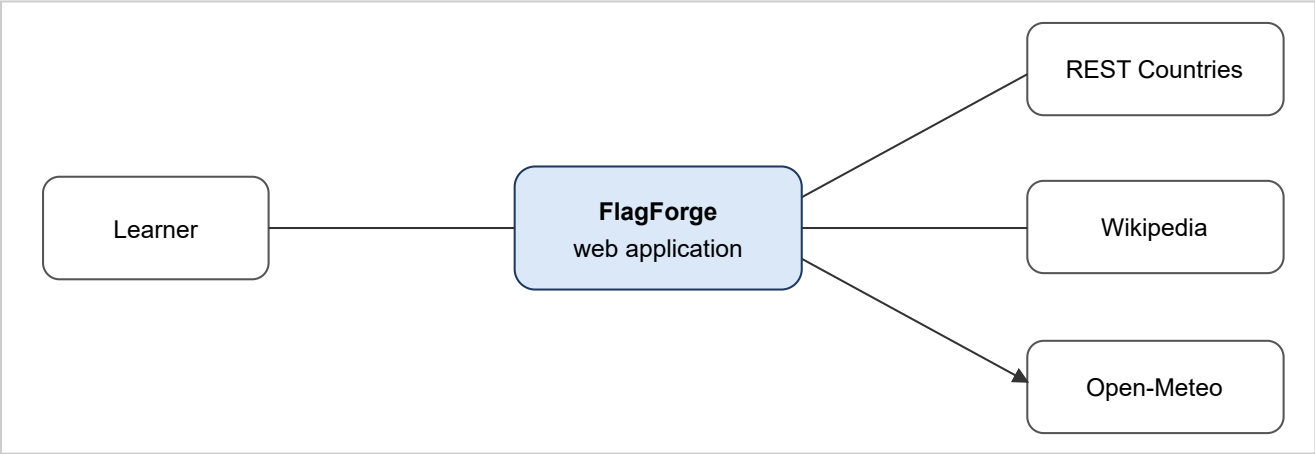


Figure 1: FlagForge between the learner and three external services

2.1 Theory Behind the Project

FlagForge is based on the principles of modern web application architecture, schema-validated data aggregation, deterministic pseudo-random generation, offline-first caching and accessible interface design.

1. Web Application Architecture

The Next.js App Router organizes pages; React renders interactive elements; TypeScript's strict typing catches incorrect data shapes before deployment. Pages such as country profiles are pre-rendered statically

for speed.

2. Validation and Normalization

The REST Countries API limits requests to 10 fields, so data is fetched in two parallel batches (identity and demographics; geography and borders). Each response is validated by a Zod schema and then merged by `normalizeCountry()` into one `Country` model used everywhere in the UI.

3. Deterministic Pseudo-Random Generation

The daily challenge must be identical for all players. The date string (for example 2026-10-24) is converted to a 32-bit integer seed, and a Mulberry32 pseudo-random number generator drives a Fisher-Yates shuffle and question selection. The same seed always produces the same quiz, without any server.

4. Gamified Scoring

Points for a correct answer are `(100 + remaining seconds x 10) x multiplier`. The multiplier rises with consecutive correct answers: 1.0x, 1.5x, 2.0x, 2.5x and a cap of 3.0x from a streak of five. A wrong answer scores zero and resets the streak.

5. Offline-First Caching

A service worker applies a strategy per resource type: cache-first for flag images, stale-while-revalidate for API data and network-first with cache fallback for pages. TanStack Query keeps fetched data fresh for 24 hours in memory.

6. Fault Tolerance

External services can be slow or fail. Every request has a timeout, failed requests are retried with exponential backoff, results are cached, and an embedded dataset provides a last-resort fallback.

2.2 Problem Definition

Country information is spread across many websites, and most learning tools either test or inform but rarely both. Learners who want to practise flags, capitals and borders, track their progress and compare countries must switch between several sites, many of which need an account or a constant connection.

External APIs also behave unpredictably: they return inconsistent fields, impose request limits and sometimes fail. A simple client that calls them directly would show errors or empty pages whenever one service has a problem.

The main problem this project addresses is: *to provide one coherent, accessible, offline-capable application that lets learners explore, practise and compare world geography while keeping their progress private on their own device.*

To be successful, the solution must:

- cover all countries with correct flags, capitals, borders and demographic data;
- offer several quiz types with adjustable difficulty and meaningful scoring;

- work without registration and keep progress between sessions;
- handle API failures gracefully and keep working offline;
- be usable by keyboard and screen reader, and on phones as well as desktops.

Difficulty	Design response	Where
Data split across services	Typed API clients, Zod validation, one Country model	src/lib/api
Inconsistent or failing APIs	Timeout, retry with backoff, cache, embedded fallback	restcountries.ts, fallback-countries.ts
Same quiz for everyone each day	Seeded PRNG from UTC date	quiz/rng.ts
Progress lost between visits	Persisted Zustand stores with versions	src/store
No connection	Service worker and offline page	public/sw.js
Hard to use without a mouse	Keyboard shortcuts, skip link, focus rings, axe tests	e2e/accessibility.spec.ts

2.3 Need for the Project

Students increasingly learn on mobile devices and expect fast, clear navigation. A browser-based atlas and quiz needs no installation and replaces printed atlases for quick revision.

Active learning. Retrieval practice, such as answering questions under mild time pressure, improves memory more than re-reading. Timed quizzes, streaks and the daily challenge make such practice a routine.

Targeted revision. Difficulty levels and the weakest-countries list let the learner concentrate on what they miss most.

Comparison. Seeing population, area, languages and currencies of up to three countries together makes differences obvious.

Privacy and optional accounts. Progress, favorites and settings are stored only in the browser; no registration or tracking is required.

Accessibility. Full keyboard operation, skip links, high-contrast and reduced-motion settings allow more learners to use the application.

User group	What it gains
Student	Revision tool, daily practice, progress tracking
Teacher	Classroom-friendly daily challenge; identical questions for everyone
Casual player	Quick games, achievements and country facts
Maintainer	Small, typed modules with unit tests

3. Literature Survey

Geography learning applications commonly combine country reference data, quiz mechanics and progress tracking. FlagForge builds on three public services, each with a distinct role.

REST Countries (v3.1) supplies structured data for sovereign states and territories: names, ISO codes, capitals, region and subregion, population, area, languages, currencies, borders, coordinates and flag image links. It is the primary data source.

Wikipedia REST API provides short article summaries, thumbnails and canonical links, used to enrich country profile pages.

Open-Meteo is a free weather forecast API that requires no key. FlagForge requests current conditions at a capital city's coordinates and translates the WMO weather code into a readable label.

A central challenge in aggregating public APIs is that responses differ in shape, may omit fields and can fail independently. FlagForge answers this with schema validation (Zod), normalization into one type, retries with exponential backoff, caching and an embedded dataset.

Gamification research suggests that streaks, short feedback loops and visible progress encourage regular practice. FlagForge applies streak multipliers, speed bonuses, achievements and a calendar heatmap. Interface design follows the Web Content Accessibility Guidelines (WCAG 2.2) and is checked with the axe-core engine.

Service	Main role	Data used	Module
REST Countries	Country reference data	Names, codes, capitals, borders, population, area, languages, currencies, flags	restcountries.ts
Wikipedia	Descriptive extract	Summary text, thumbnail, page URL	wikipedia.ts
Open-Meteo	Capital weather	Temperature, wind, WMO weather code, day/night	weather.ts
FlagCDN	Flag images	PNG/SVG flags (cached by service worker)	FlagImage component

4. Analysis & Design

The analysis phase identifies the learners who use the system, the problems they face, and the functional and non-functional requirements.

Functional analysis. The system must provide searchable and filterable country exploration; six quiz modes with difficulty, timer, scoring and review; a daily challenge; a three-country comparison; a statistics dashboard; and country profile pages with weather and encyclopedia content.

Non-functional analysis. The system must be responsive, keyboard accessible, privacy-preserving, installable and resilient to unavailable external APIs. It must communicate errors instead of silently hiding them.

Requirement gathering. Requirements follow learner journeys, such as finding a country, practising for five minutes, comparing two nations, and checking progress over a week.

System architecture. A modular, layered web architecture. Route files, API clients, the quiz engine, hooks, persisted stores and presentational components are kept in separate folders, which supports testing and maintainability.

Data design. No server database is used. Country data is fetched and cached; the user's progress, favorites and settings live in three versioned localStorage stores.

User interface design. A hero page with direct actions, a sticky navigation bar with command palette, a mobile tab bar, card and list layouts, light, dark and high-contrast themes.

Security and validation design. A strict Content-Security-Policy, `X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, Zod validation of all external data and no storage of personal information.

ID	Requirement	Type
R1	Search and filter all countries by name, capital, region, population	Functional
R2	Play quizzes with six modes, three difficulty levels, scoring and review	Functional
R3	Provide a deterministic daily challenge	Functional
R4	Compare up to three countries with charts	Functional
R5	Persist progress, favorites and settings locally	Functional
R6	Survive slow, failing or absent network	Non-functional
R7	Be accessible by keyboard and meet WCAG 2.2 AA	Non-functional
R8	Protect users with strict security headers and local-only data	Non-functional

4.1 Process Model

The project follows a phased software-development process in which each phase feeds the next and findings flow back when needed.

- **Requirement analysis:** identify learner journeys; list functional and non-functional requirements.
- **System design:** define routes, the Country model, API clients, store schemas and the quiz engine interface.
- **Implementation:** build Next.js pages, reusable React components and small TypeScript modules.
- **Testing:** unit and component tests with Vitest; browser and accessibility tests with Playwright and axe-core.
- **Deployment and maintenance:** production build, CI pipeline, deployment to a static/Node host or Docker, monitoring of upstream APIs.

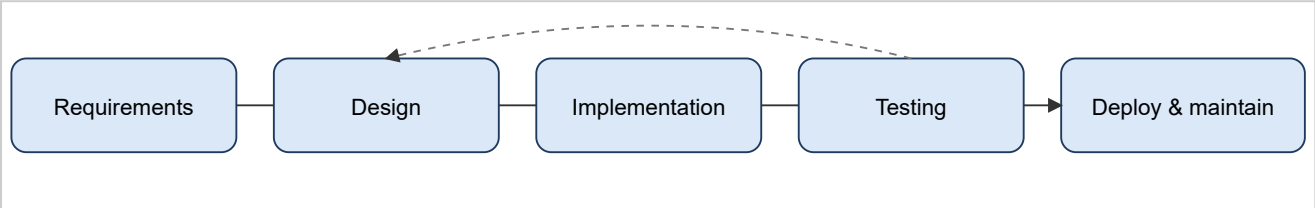


Figure 2: Development process model (dashed arrow: feedback to design)

Phase	Main output	Checked by
Requirement analysis	Journeys, requirement list	Review
System design	Types, diagrams, store schemas	Design walkthrough
Implementation	Next.js modules	tsc (strict), ESLint
Testing	Unit, browser and a11y tests	Vitest, Playwright, axe-core
Deployment	Production build, CI workflow	next build, GitHub Actions

4.2 Flow of Project

The flow follows the learner's journey. On the home page the learner picks Play Quiz, Explore Countries or Compare Nations. The data layer loads countries (cached for 24 hours in memory and by the service worker). In the explorer the learner filters and opens a country profile; in the arena the learner picks a mode and difficulty, answers questions and sees a review; results update the persisted progress store, which feeds the stats dashboard and achievements.

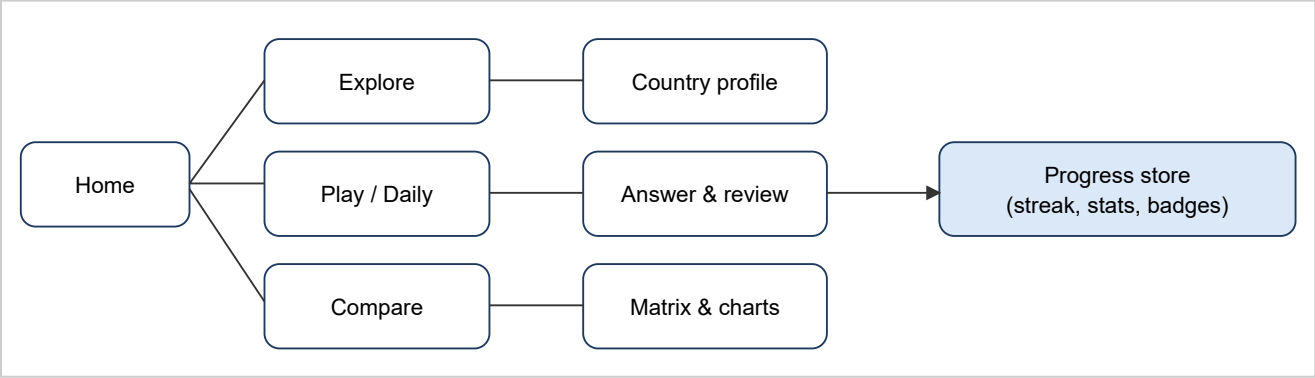


Figure 3: Flow of the project

Quiz path: configure mode, difficulty, number of questions and timer; the engine builds the questions; each answer is scored; the game-over screen shows the review; the progress store records answers, game end and streak.

Daily path: the UTC date becomes the seed; ten flag questions are generated; one result per day is recorded and can be shared as an emoji grid.

Offline path: if the network fails, cached responses or the embedded 30-country dataset are used, and visited pages open from the service-worker cache, with `/offline.html` as a last resort.

4.3 Feasibility Study

Technical feasibility. The application relies on well-documented technologies (Next.js, React, TypeScript, Tailwind CSS) and keyless public APIs. It runs in any modern browser. API limits and availability are the main constraints and are handled by timeouts, retries, caching and fallback data.

Economic feasibility. All development tools and packages are free and open source. A static or small Node deployment fits free hosting tiers. No database or paid API key is needed.

Operational feasibility. Learners can use the application immediately without registering. The interface is responsive and keyboard accessible.

Schedule feasibility. Modules (data layer, quiz engine, explorer, compare, stats) can be built and tested incrementally.

Risk	Effect	Response in the design
API downtime	Missing data	Retry with backoff; embedded 30-country fallback
Changed API shape	Runtime errors	Zod validation at the boundary; typed AppError
Slow network	Poor experience	Timeouts, caches, skeleton loaders
Browser storage cleared	Progress lost	Documented; versioned stores with safe defaults
Accessibility regressions	Excluded users	axe-core checks in Playwright and CI

Dimension	Verdict	Basis
Technical	Feasible	Documented frameworks and public APIs
Economic	Feasible	Free tools and hosting tier; no keys
Operational	Feasible	No account needed; responsive UI
Schedule	Feasible	Incremental modules over eight weeks

4.4 Cost Analysis

The following figures are qualitative planning estimates, not actual project expenses.

Cost item	Development	Demo deployment	Larger service
Tools (VS Code, Node.js, pnpm, packages)	No licence fees	No licence fees	No licence fees
Hosting	Local machine	Free tier, within limits	Paid plan as traffic grows
Database	Not needed	Not needed	Needed only if accounts are added
API requests	Free, keyless	Free, keyless	Fair-use limits; possible self-hosted mirror
Domain name	Not needed	Optional	Yearly fee
Effort	Main cost: student time for design, coding, testing and documentation		

Because all user data stays in the browser, the system has almost no operating cost per user. The benefits are mostly qualitative: easier revision, a regular practice routine and quick country comparisons. A formal return-on-investment calculation would require measured usage, which is not available.

Cost driver	Why it can grow	How the design limits it
API requests	Every visitor loads country data	24 h query cache, 10-minute client cache, service-worker cache
Hosting traffic	More page views	Static pre-rendering, immutable assets
Maintenance	APIs change	Isolated clients, one per service; Zod catches changes early
Testing effort	Many browsers and screens	Automated unit, E2E and accessibility tests

4.5 Data Flow Diagram (Level 0)

A data flow diagram (DFD) shows how data moves between external actors, processes and data stores. The context diagram treats FlagForge as a single process.

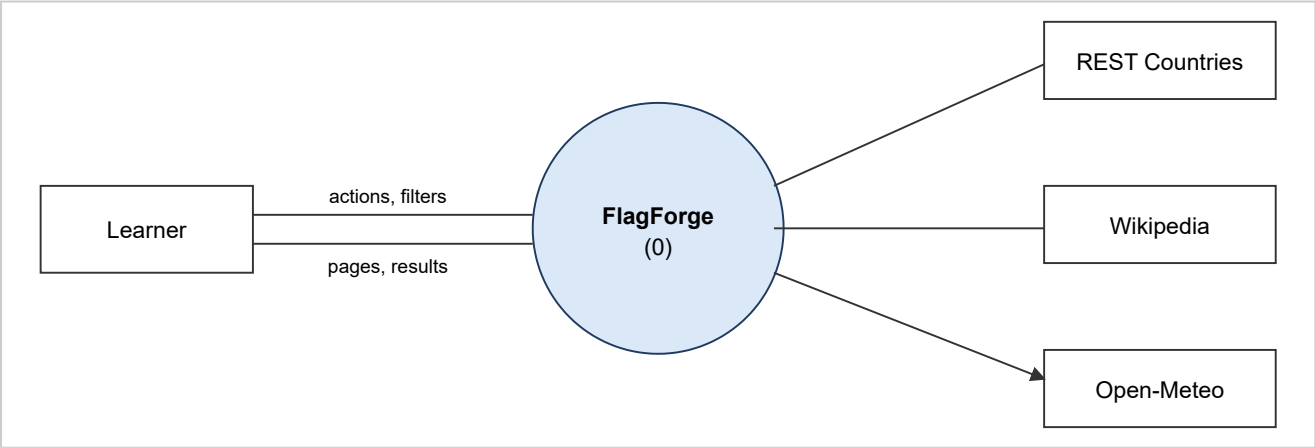


Figure 4: DFD Level 0 (context diagram)

Data flow	Source	Destination	Content
Action	Learner	Web app	Search, filter, answers, settings
Pages and results	Web app	Learner	Country lists, questions, scores, charts
Country request	Web app	REST Countries	Two field batches
Summary request	Web app	Wikipedia	Article title
Weather request	Web app	Open-Meteo	Capital latitude and longitude

Data store	Holds	Lifetime
localStorage: flagforge-progress (v2)	Streak, accuracy, history, daily results, achievements	Until browser data is cleared
localStorage: flagforge-favorites (v1)	Favorite country codes	Until cleared
localStorage: flagforge-settings (v2)	Theme, sound, motion, units, contrast	Until cleared
Query cache and service-worker caches	Country, Wikipedia, weather data; flags; pages	24 h stale time; entry limits

4.5.1 Data Flow Diagram (Level 1)

At Level 1 the application is split into four processes: data loading (fetch, validate, normalize), exploration (search and filter), quiz handling (generate, score, review) and progress handling (update persisted stores).

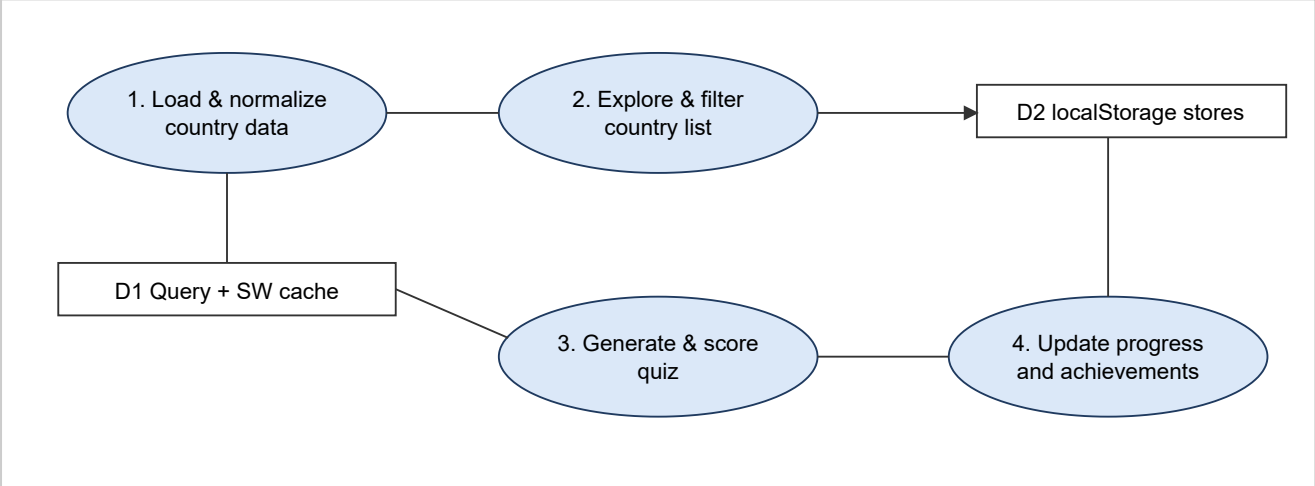


Figure 5: DFD Level 1

Process	Input	Output	Data store
1. Load and normalize	API responses	Country[]	Query cache, SW cache
2. Explore and filter	Query, filters	Filtered list	Favorites store
3. Generate and score	Mode, difficulty, seed	Questions, points	None (pure functions)
4. Update progress	Answers, game end	Streaks, stats, badges	Progress store

Failure paths

Situation	Reaction
Request times out (8 s countries, 6 s weather/wiki)	Up to 2 retries with exponential backoff
404 or validation error	Not retried; typed AppError shown in an error state
Country API unreachable	Embedded 30-country dataset is used
Weather or Wikipedia unavailable	The card shows an error state; the rest of the page still works
Browser offline	Service worker serves cached pages or /offline.html

4.6 UML Diagrams

The Unified Modeling Language (UML) provides standard diagrams for describing structure and behavior. The diagrams below are conceptual models; exact fields should be checked against the TypeScript types in `src/types`.

Diagram	What it shows in FlagForge
4.6.1 Use case	The learner's functions
4.6.2 Class	Country, Question, store state types
4.6.3 Activity	A quiz session from configuration to review
4.6.4 Sequence	Messages while loading countries
4.6.5 Data model	Persisted stores and their contents (no server database)
4.6.6 Component	UI, hooks, stores, API clients and services

4.6.1 Use Case Diagram

The system has one principal actor, the learner, supported by external services. All functions are available without signing in.

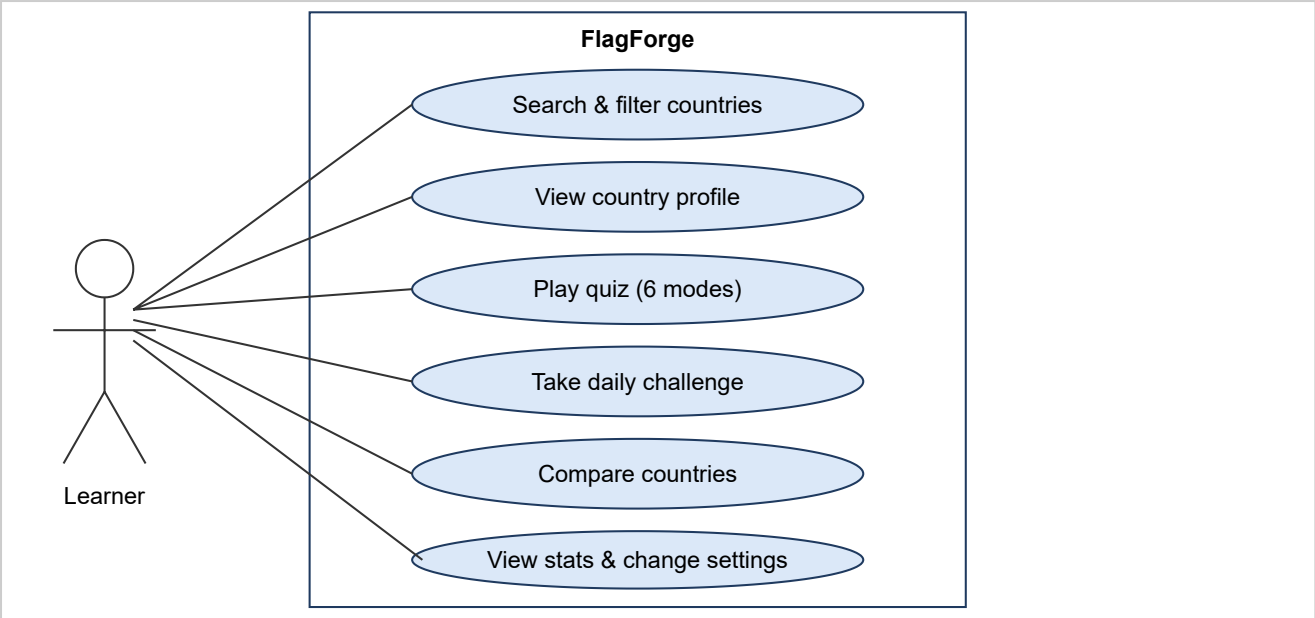


Figure 6: Use case diagram

ID	Use case	Outcome
UC1	Search and filter countries	Filtered grid or list; favorites
UC2	View country profile	Facts, borders, map, weather, Wikipedia summary
UC3	Play a quiz	Score, streak, review screen
UC4	Take the daily challenge	Shared 10-question result; shareable grid
UC5	Compare countries	Matrix and charts for up to 3 countries
UC6	View stats	Heatmap, radar, weakest countries, badges
UC7	Change settings	Theme, sound, motion, contrast, units

4.6.2 Class Diagram

A `Country` carries normalized descriptive fields. A `Question` references a target country and a list of options; exactly one option is correct. The progress store aggregates answers into accuracy and streak data.

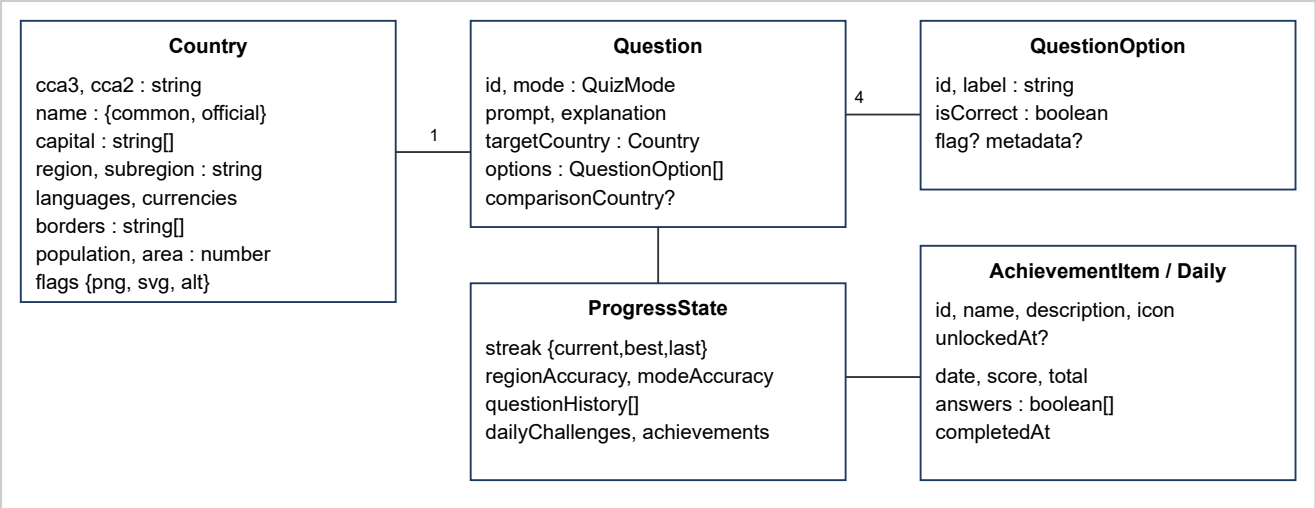


Figure 7: Class diagram

Example object: a question of mode `flag_to_country` whose target is France has four options (France plus three distractors from the same region where possible), and exactly one option has `isCorrect = true`.

Type	Attribute	Description
Country	cca3 / cca2	ISO 3166-1 alpha-3 and alpha-2 codes
Country	region, subregion, borders	Used by filters, difficulty and border questions
Question	mode	flag_to_country, country_to_capital, capital_to_country, higher_lower, which_borders or map_location
Question	metadata	Metric (population or area) and values for higher/lower; border country codes
StreakData	current, best, lastActiveDate	Day streak of activity

4.6.3 Activity Diagram

The primary activity begins when the learner opens the Quiz Arena. The learner chooses mode, difficulty and options; the engine generates questions; for every question the learner answers (or the timer expires), the score is calculated and feedback is shown; at the end the review is displayed and progress is saved.

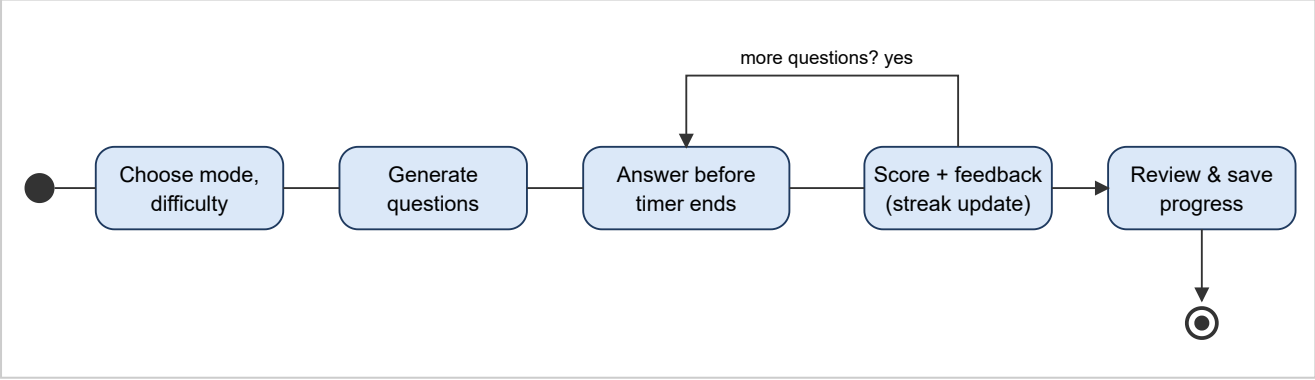


Figure 8: Activity diagram of a quiz session

The quiz can be paused with Esc; unlimited-time mode removes the timer for learners who need more time (WCAG 2.2.2).

4.6.4 Sequence Diagram

When a view needs countries, a hook asks TanStack Query. On a cache miss the REST Countries client requests the two field batches in parallel, validates and merges them, and returns the typed result. If the request fails after retries, the embedded dataset is returned.

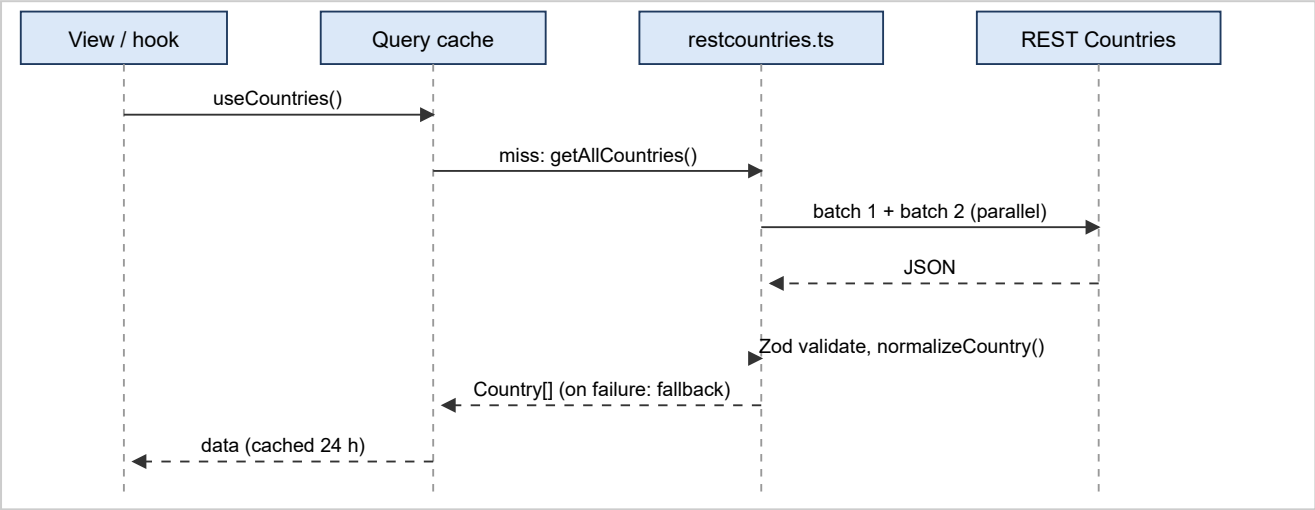


Figure 9: Sequence diagram for loading countries

No.	From to	Message
1	Hook to query cache	useCountries() requests the list
2	Cache to client	On a miss, getAllCountries()
3	Client to API	Two parallel requests with 8 s timeout and up to 2 retries
4	Client	Validate with Zod, merge, normalize
5	Client to cache	Typed Country[] (or the embedded fallback)
6	Cache to view	Render with data; stays fresh for 24 hours

4.6.5 Data Model (ER-equivalent)

FlagForge has no server database. Its persistent entities are three versioned localStorage documents managed by Zustand's `persist` middleware. If the stored version is older, a migration step fills in new fields with defaults.



Figure 10: Persisted data model (three localStorage stores)

Entries in the progress and favorites stores refer to countries by their `cca3` code, so no country data is duplicated in storage. The exact field list of the progress store is defined in `src/types/store.ts`.

4.6.6 Component Diagram

React components call hooks; hooks read the TanStack Query cache and Zustand stores; the data layer talks to the external services. Dependencies run in one direction, which keeps the quiz engine and API clients testable without a browser.

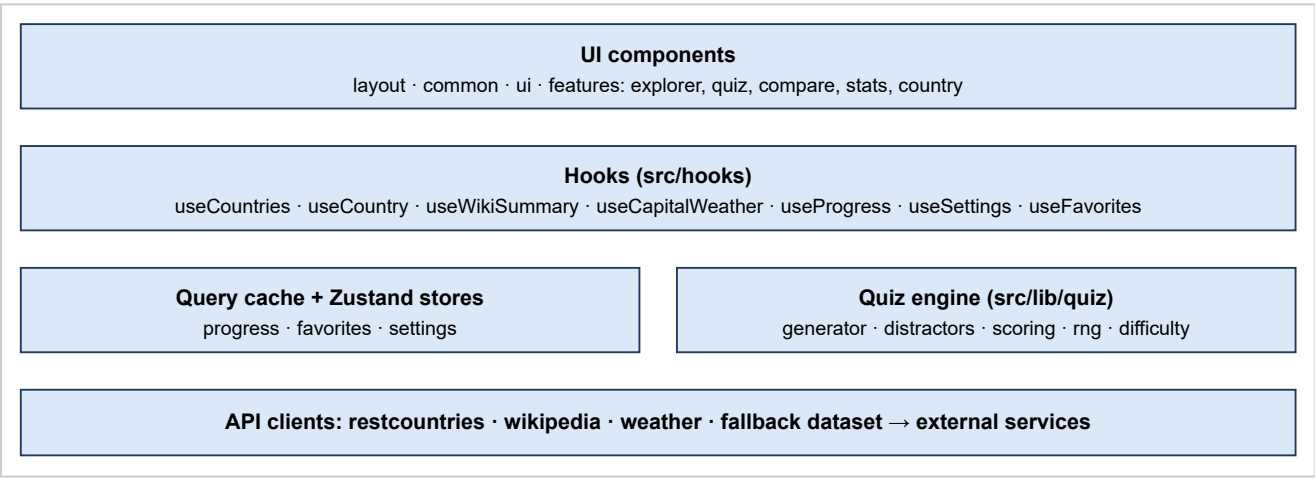


Figure 11: Component diagram

The code base is divided into folders with clear responsibilities (counts from the repository at the time of writing).

Folder	Files	Lines	Responsibility
src/app	16	3,443	Pages, metadata, manifest, sitemap, OG image
src/components	70	7,210	UI primitives, layout and feature components
src/lib	17	3,029	API clients, quiz engine, utilities, query client
src/hooks	8	391	Public hooks API
src/store	4	487	Persisted Zustand stores
src/tests	5	1,345	Vitest unit and component tests
e2e	4	270	Playwright specifications

Three rules keep the structure clean: components may import hooks and libraries but never the reverse; one client per external service; one place per decision (scoring rules only in `scoring.ts`, difficulty only in `difficulty.ts`).

4.7 Gantt Chart

A Gantt chart is a bar chart used in project management to show the timeline of tasks. The proposed eight-week academic plan is shown below; it is a plan, not a verified log of actual hours.

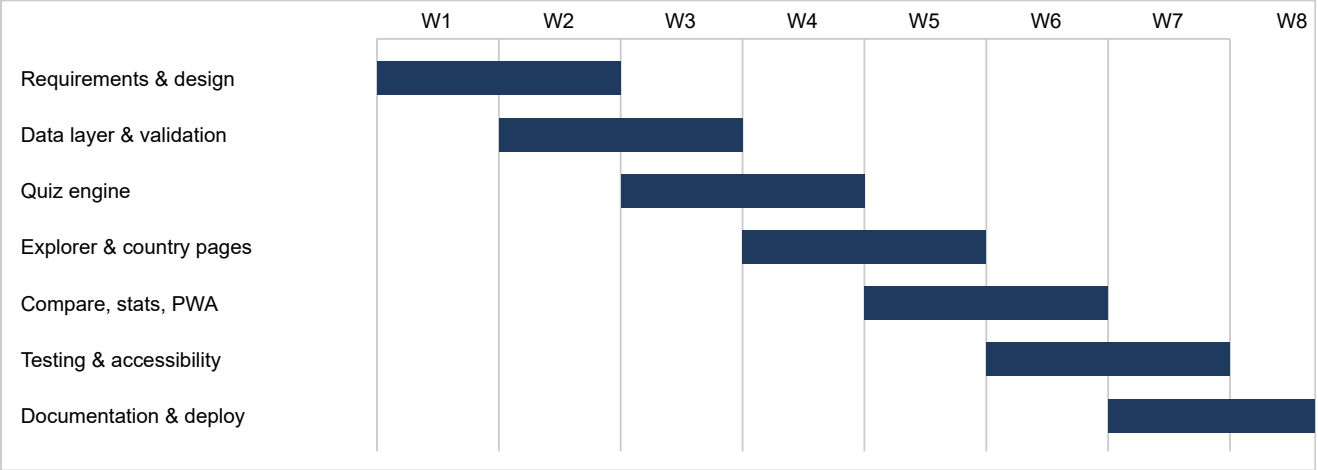


Figure 12: Project development schedule (proposed)

Weeks	Phase	Deliverable
1 - 2	Requirements and design	Requirement list, diagrams, types
2 - 3	Data layer	API clients, Zod schemas, fallback dataset
3 - 4	Quiz engine	Generators, scoring, seeded daily challenge
4 - 5	Explorer and country pages	Filters, search, profiles
5 - 6	Compare, stats and PWA	Charts, achievements, service worker
6 - 7	Testing and accessibility	Vitest, Playwright, axe-core
7 - 8	Documentation and deployment	Report, CI, production build

4.8 Technologies Used

Versions are taken from the project's `package.json`.

Area	Technology	Usage in the project
Framework	Next.js 15.2 (App Router), React 19	Routing, pre-rendering, interactive components
Language	TypeScript (strict)	Shared Country, Question and store types
Styling	Tailwind CSS 4, class-variance-authority, tailwind-merge	Responsive layout, themes, variants
UI primitives	Radix UI (dialog, dropdown, select, slider, tabs, tooltip, progress)	Accessible dialogs, menus and controls
Server state	TanStack Query 5 (+ persist client)	Fetching, caching, retries
Client state	Zustand 5 with persist middleware	Progress, favorites, settings
Validation	Zod 3	API response schemas
Search	Fuse.js 7	Fuzzy country search
Charts and motion	Recharts 2, Framer Motion 12, canvas-confetti, lucide-react	Compare and stats charts, transitions, effects, icons
Unit testing	Vitest, Testing Library, jsdom	Data layer, stores, quiz engine, explorer
Browser testing	Playwright, @axe-core/playwright	End-to-end and accessibility checks
Quality tooling	ESLint 9, Prettier, tsc, GitHub Actions, pnpm	Linting, formatting, type checking, CI

The stack keeps the application a single TypeScript codebase: pages, data access, quiz logic and tests share one set of types.

4.9 System Architecture

The browser communicates with Next.js pages; typed API clients fetch and validate external data; the TanStack Query cache and persisted stores hold state; a service worker adds offline capability; feature components render the result.

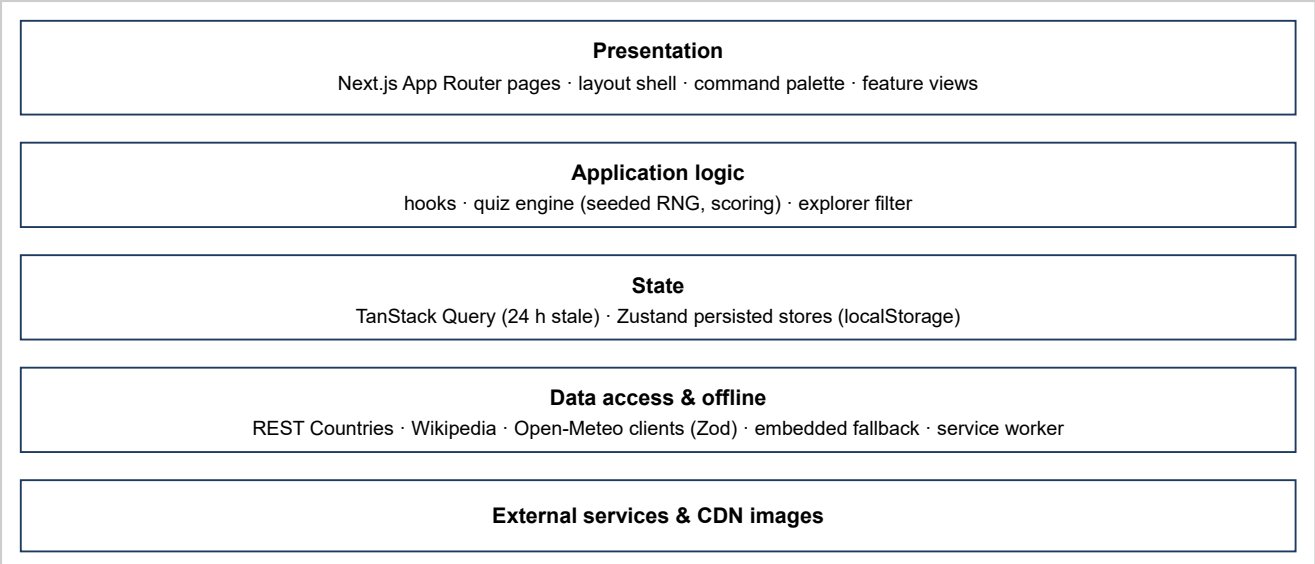


Figure 13: Layered system architecture

Setting	Value
Country request timeout / retries	8 s / 2 retries with exponential backoff
Weather and Wikipedia timeout / retries	6 s / 2 retries
In-memory client cache (countries)	10 minutes
Weather cache	15 minutes
TanStack Query staleTime / gcTime	24 hours / 48 hours
Service worker caches	static, flags (max 300), data (max 100), pages
Offline dataset	30 countries embedded

Security headers (next.config.ts)

- Content-Security-Policy restricting scripts, images (FlagCDN, Wikimedia) and connections (REST Countries, Wikipedia, Open-Meteo) to known origins; `frame-ancestors 'none'` and `object-src 'none'`.
- `X-Frame-Options: DENY` and `X-Content-Type-Options: nosniff`.

This is a resilience measure and not a guarantee of uninterrupted service.

5. Implementation

FlagForge is implemented as a single Next.js project. Routes live under `src/app`, UI components under `src/components`, and logic under `src/lib`, `src/hooks` and `src/store`.

Folder	Responsibility
src/app	Pages: home, explore, play, daily, compare, stats, country/[code], design; error and not-found pages; manifest, robots, sitemap, Open Graph image
src/components	ui (primitives), common (flag image, chips, stat card), layout (navbar, tab bar, command palette, PWA prompt), features (explorer, quiz, compare, stats, country)
src/lib/api	REST Countries, Wikipedia and Open-Meteo clients; embedded fallback dataset
src/lib/quiz	Question generators, distractors, difficulty, scoring, seeded RNG
src/hooks	Public hooks that combine queries and stores
src/store	Versioned, persisted Zustand stores
public	Service worker (sw.js), offline page, icons
src/tests, e2e	Vitest and Playwright tests

Key design decisions. All external data is validated at the boundary and normalized once. The quiz engine consists of pure functions that take countries and a random function, so the daily challenge is reproducible and unit-testable. Scoring and difficulty rules each live in a single file. Stores are versioned so that future changes can migrate old data. A service worker is used instead of a heavy PWA plugin to keep caching rules explicit.

5.1 Code

The listings below are taken from the repository (github.com/KAUSHALCODER123/geo-guess).

src/lib/quiz/scoring.ts - streak multiplier and score

```
export function calculateStreakMultiplier(streak: number): number {
  if (streak <= 1) return 1.0;
  if (streak === 2) return 1.5;
  if (streak === 3) return 2.0;
  if (streak === 4) return 2.5;
  return 3.0; // 5 or higher capped at 3.0x
}

export function calculateScore(input: ScoreCalculationInput): ScoreResult {
  const { isCorrect, timeRemaining, streak, basePoints = 100 } = input;
  if (!isCorrect) {
    return { pointsEarned: 0, speedBonus: 0, multiplier: 1.0, newStreak: 0 };
  }
  const newStreak = streak + 1;
  const multiplier = calculateStreakMultiplier(newStreak);
  const remainingSeconds = Math.max(0, Math.floor(timeRemaining));
  const speedBonus = remainingSeconds * 10;
  const pointsEarned = Math.round((basePoints + speedBonus) * multiplier);
  return { pointsEarned, speedBonus, multiplier, newStreak };
}
```

src/lib/quiz/rng.ts - seeded generator for the daily challenge

```
export function mulberry32(seed: number): () => number {
  let s = seed | 0;
  return function (): number {
    s = (s + 0x6d2b79f5) | 0;
    let t = Math.imul(s ^ (s >>> 15), 1 | s);
    t = (t + Math.imul(t ^ (t >>> 7), 61 | t)) ^ t;
    return ((t ^ (t >>> 14)) >>> 0) / 4294967296;
  };
}

export function stringToSeed(str: string): number {   // e.g. "2026-10-24"
  let hash = 0;
  for (let i = 0; i < str.length; i++) {
    hash = (hash << 5) - hash + str.charCodeAt(i);
    hash |= 0;
  }
  return hash;
}

export function shuffle<T>(array: readonly T[], rng: () => number = Math.random): T[] {
  const result = [...array];
  for (let i = result.length - 1; i > 0; i--) {
    const j = Math.floor(rng() * (i + 1));
    [result[i], result[j]] = [result[j], result[i]];
  }
  return result;
}
```

src/lib/quiz/difficulty.ts - difficulty classification

```
export function getCountryDifficulty(country: Country): "easy" | "medium" | "hard" {
  if (country.population > 25_000_000 || ICONIC_CCA3.has(country.cca3)) return "easy";
  if (country.population >= 5_000_000 && country.population <= 25_000_000) return "medium";
  return "hard";
}

export function filterByDifficulty(countries: Country[], difficulty: Difficulty): Country[] {
  if (difficulty === "all") return countries;
  const filtered = countries.filter((c) => getCountryDifficulty(c) === difficulty);
  // too few countries for meaningful questions: fall back to all
  return filtered.length >= 4 ? filtered : countries;
}
```

src/lib/query-client.ts - cache and retry policy

```
export function createQueryClient(): QueryClient {
  return new QueryClient({
    defaultOptions: {
      queries: {
        staleTime: 24 * 60 * 60 * 1000,   // 24 hours
        gcTime: 48 * 60 * 60 * 1000,      // 48 hours
        retry: (failureCount, error) => {
          if (error && typeof error === "object" && "isNotFound" in error &&
              (error as { isNotFound: boolean }).isNotFound) return false; // never retry 404
          return failureCount < 2;
        },
        refetchOnWindowFocus: false,
        refetchOnReconnect: true,
      },
      mutations: { retry: 1 },
    },
  });
}
```

src/lib/api/restcountries.ts - constants of the HTTP layer

```
const DEFAULT_TIMEOUT_MS = 8000;
const MAX_RETRIES = 2;
const CACHE_TTL_MS = 10 * 60 * 1000; // 10 minutes
// fetchWithRetry(): AbortController timeout, exponential backoff between attempts,
// 404 and other non-retryable errors are thrown immediately as a typed AppError.
// getAllCountries(): batch 1 and batch 2 are fetched in parallel, validated with Zod,
// merged by cca3, and the embedded fallback dataset is returned if the network fails.
```

next.config.ts - Content-Security-Policy

```
default-src 'self';
img-src 'self' blob: data: https://flagcdn.com https://upload.wikimedia.org
https://commons.wikimedia.org;
connect-src 'self' https://restcountries.com https://en.wikipedia.org https://api.open-
meteo.com;
frame-ancestors 'none'; object-src 'none'; base-uri 'self'; form-action 'self';
```

5.2 Graphical User Interface

The following screens are captured from the running application.

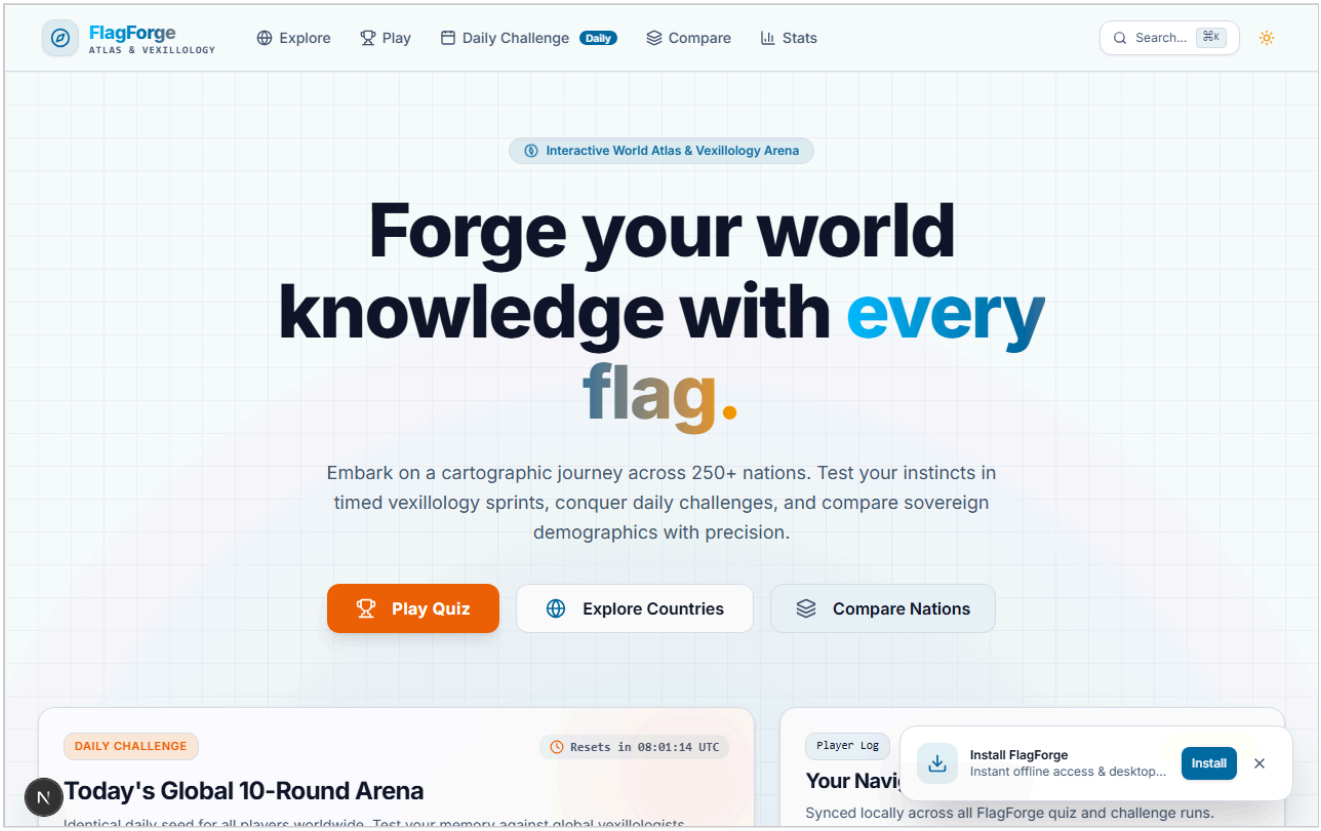


Figure 14: Home page - hero with Play, Explore and Compare actions

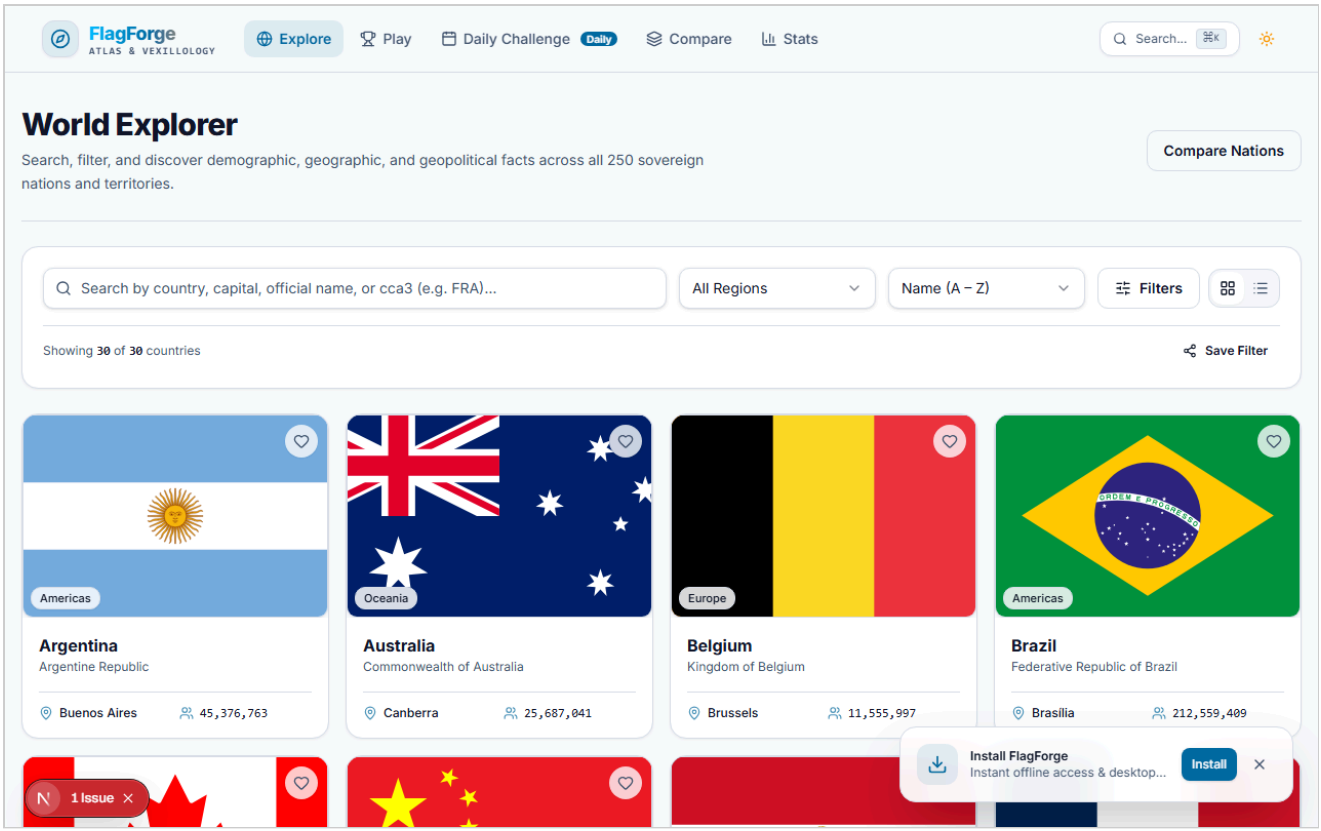


Figure 15: World Atlas Explorer with search and filters

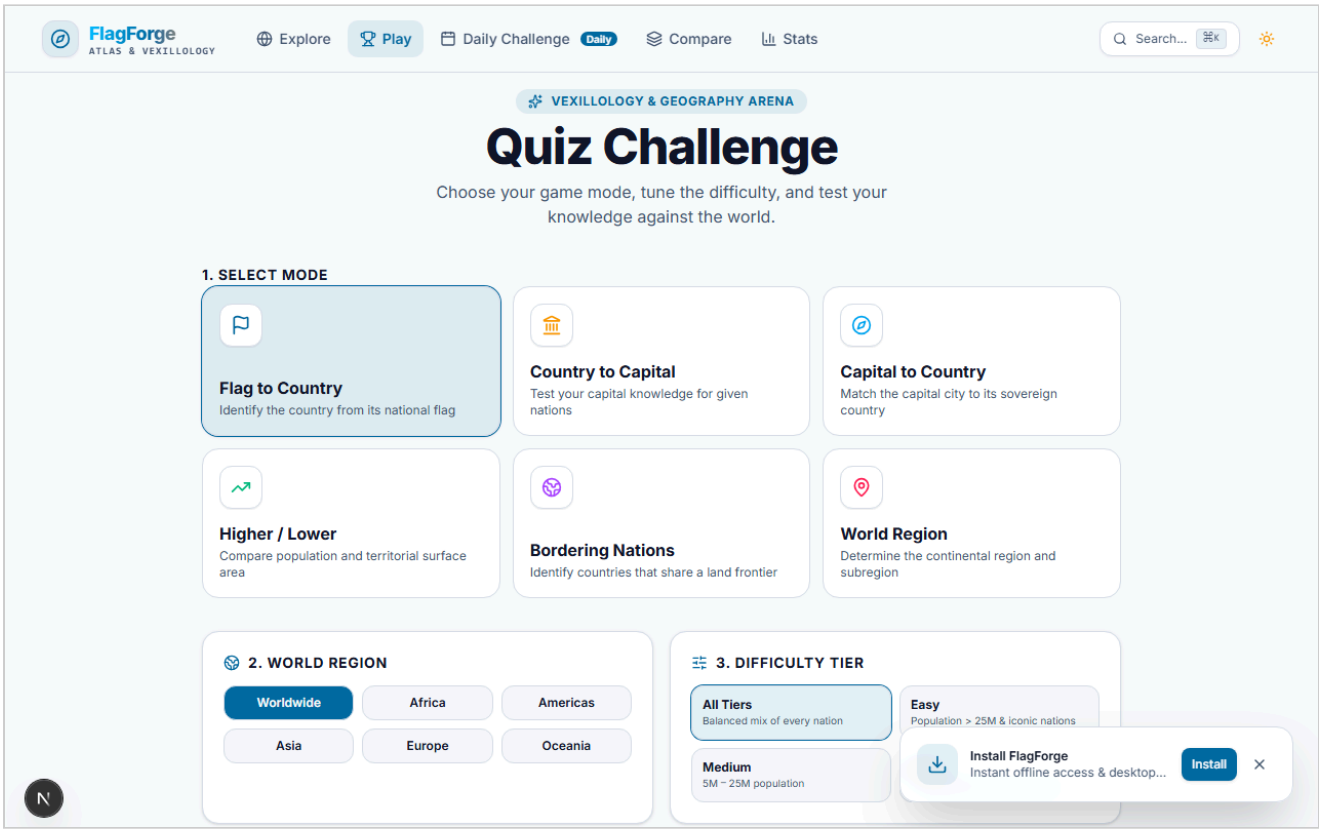


Figure 16: Quiz Arena configuration

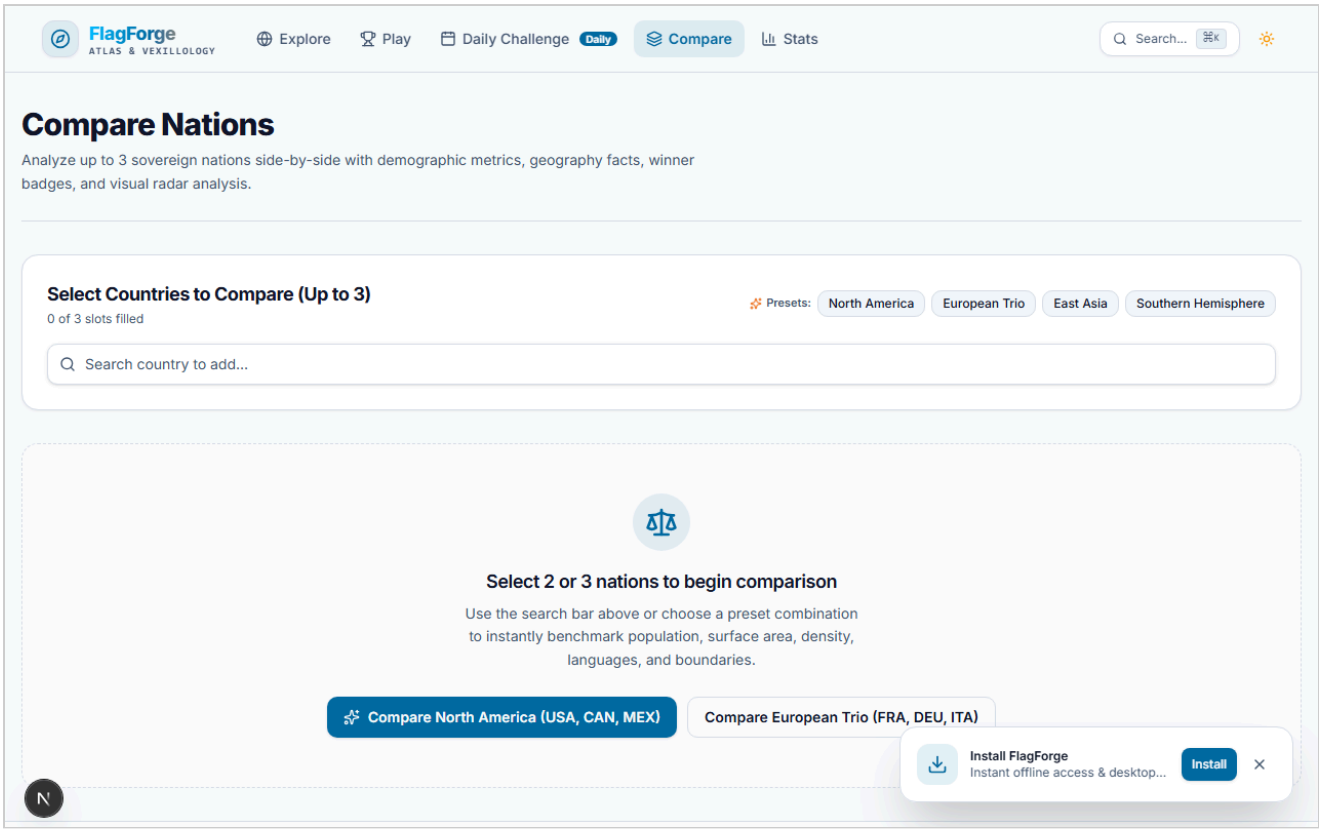


Figure 17: Compare countries

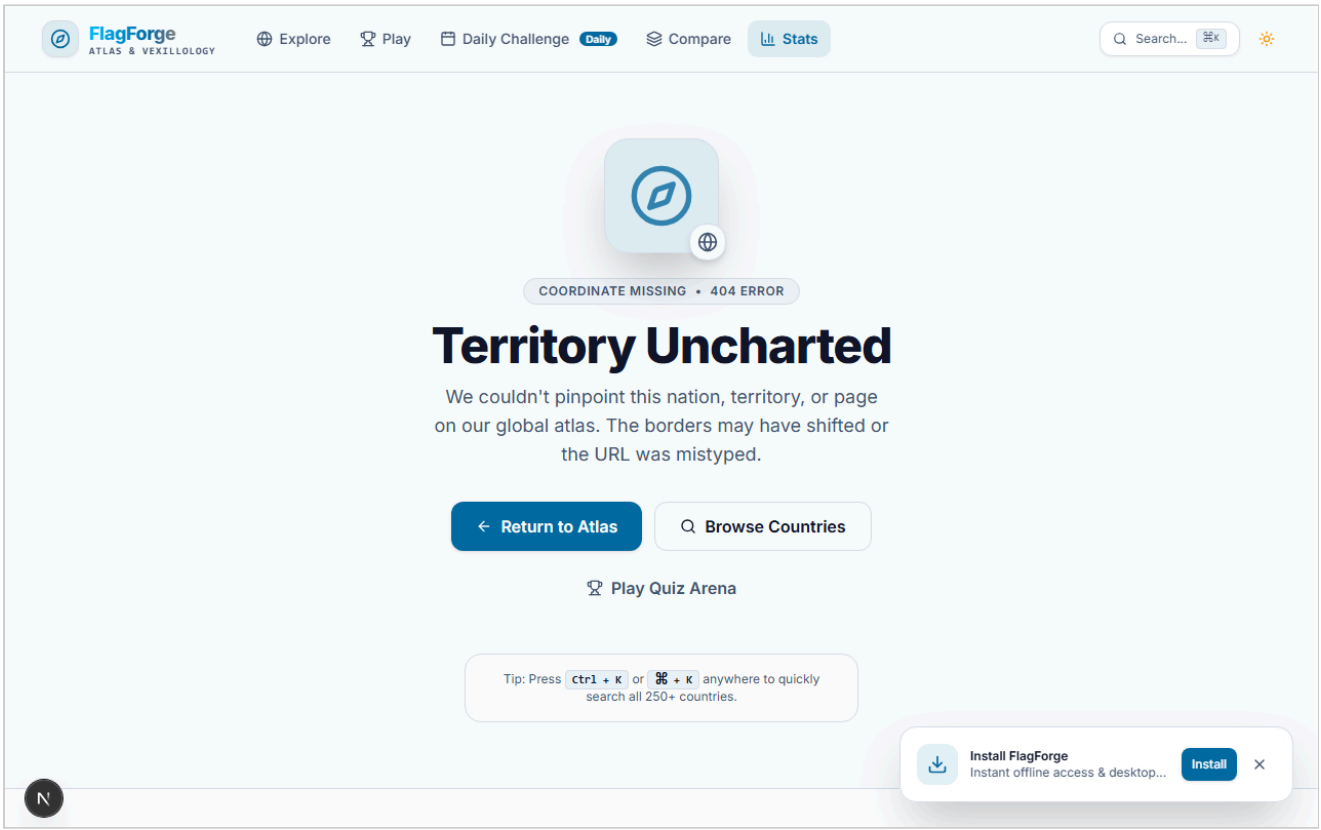


Figure 18: Statistics dashboard

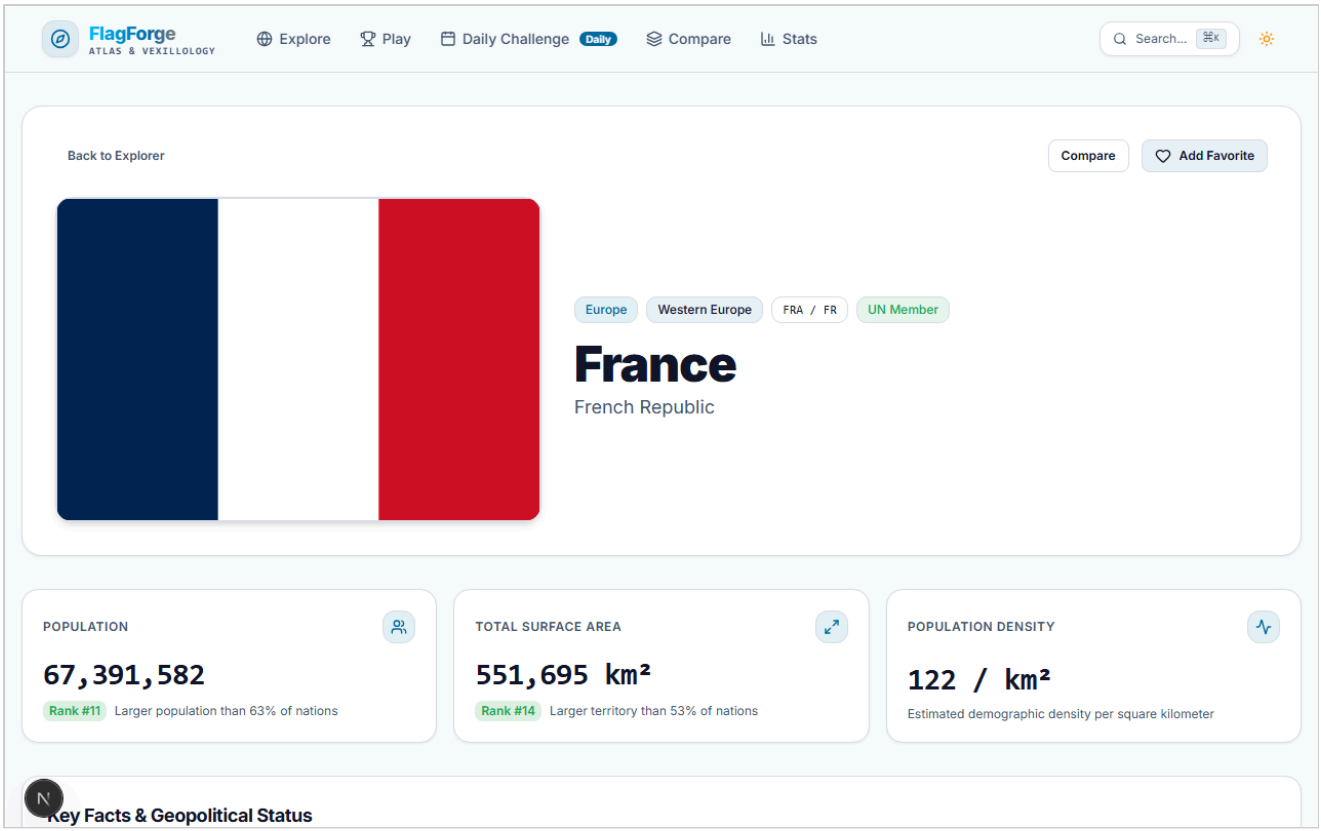


Figure 19: Country profile page

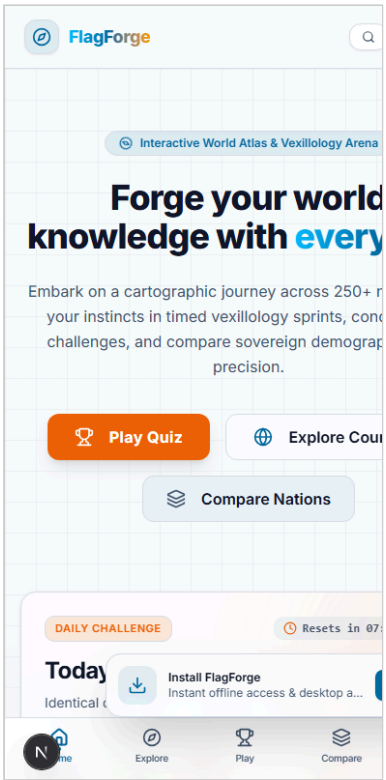


Figure 20: Home page on a phone

Area	Behaviour on a phone
Navigation	Bottom tab bar (MobileTabBar) and command palette
Explorer	Card grid collapses to one column; filters in a sheet
Quiz	Large touch-friendly answer buttons; keyboard shortcuts on desktop
Theme	System, light or dark; optional high-contrast and reduced motion
Install	PWA install prompt; offline page when disconnected

5.3 Test Cases

The repository contains Vitest tests for the data layer, stores, quiz engine and explorer, and Playwright specifications for the main journeys and accessibility. At the time of writing, running `vitest run` reported **4 test files and 58 tests, all passing**.

ID	Test scenario	Expected result	Related tests
TC01	Search countries by name or capital	Matching countries are listed	data-layer.test.ts; e2e/explorer
TC02	Filter by region	Only countries of that region are shown	explore.test.tsx; e2e/explorer
TC03	Toggle a favorite	The country is stored and displayed as a favorite	stores.test.ts; e2e/explorer
TC04	Open a country profile	Profile page with facts and borders	e2e/explorer
TC05	Generate quiz questions	Each question has 4 options, exactly one correct	quiz-engine.test.ts
TC06	Score an answer	Points follow the formula; wrong answer resets streak	quiz-engine.test.ts
TC07	Daily challenge determinism	The same date seed gives identical questions	quiz-engine.test.ts
TC08	Play a quiz with keys 1-4	Answers register; game-over screen appears	e2e/quiz
TC09	Network failure	Embedded dataset is returned	data-layer.test.ts
TC10	Compare 2 or 3 countries; remove one	Matrix and charts update	e2e/compare
TC11	Progress persists	Stores restore their state	stores.test.ts
TC12	Accessibility audit of main routes	No axe-core violations	e2e/accessibility

Test area	File	Tests
Data layer and API clients	src/tests/data-layer.test.ts	14
Explorer components	src/tests/explore.test.tsx	20
Quiz engine	src/tests/quiz-engine.test.ts	15
Stores	src/tests/stores.test.ts	9
Total (Vitest)		**58**

Commands: `pnpm run typecheck`, `pnpm run lint`, `pnpm run test`, `pnpm run e2e` and `pnpm run build`. The Vitest result above was observed when this report was prepared; Playwright and accessibility results were not re-run for this report.

6. Conclusion & Future Scope

6.1 Conclusion

FlagForge demonstrates how public country data, quiz mechanics and personal progress tracking can be combined into one browser-based, offline-capable application. It fulfils its central goal: a convenient, accessible interface for exploring, practising and comparing world geography, built on a modular architecture that is practical to extend and maintain.

The project illustrates practical engineering decisions:

- validating external data with Zod and normalizing it into one model;
- pure, seeded quiz generation that makes the daily challenge reproducible;
- graceful degradation through timeouts, retries, caching and an embedded dataset;
- local-first, versioned persistence with no account and no server database;
- a service worker with a strategy per resource type;
- a strict Content-Security-Policy and keyboard-first accessibility;
- automated unit, browser and accessibility tests.

Limitations. Progress is stored only in the current browser and is lost if site data is cleared; there is no cross-device sync or global leaderboard. The embedded offline dataset covers 30 countries, so full offline use depends on data cached earlier. The CSP currently permits inline scripts required by the framework. Data accuracy depends on third-party APIs.

Objective	Outcome
One interface for explore, practise, compare	Explorer, six-mode quiz, compare view, stats
Reliable data from several services	Typed clients, Zod validation, one Country model
Work without a connection	Service worker, caches, embedded fallback
Keep progress private	Versioned localStorage stores
Be accessible	Keyboard shortcuts, skip link, axe-core tests
Verify behaviour	58 passing Vitest tests; Playwright specs

Lessons learned. External data must be treated as untrusted; small pure functions are easy to test and explain; accessibility is cheapest when designed in from the start.

6.2 Future Scope

The following enhancements are proposed extensions, not features claimed to be implemented.

- Optional accounts with cross-device synchronization of progress.
- Global or classroom leaderboards for the daily challenge.

- An interactive world map for the map-location quiz mode and region drills.
- More content: capitals' landmarks, currencies and languages quizzes, and flag-design trivia.
- Complete offline dataset for all countries via a build-time snapshot.
- Multilingual interface and country names.
- Spaced-repetition scheduling based on the weakest-countries data.
- Tightening the Content-Security-Policy with nonces instead of inline scripts.

Extension	Module affected	Prerequisite
Accounts and sync	Stores, new backend	Authentication and database design
Leaderboards	Daily challenge, backend	Accounts, anti-cheat design
Spaced repetition	Quiz engine, progress store	Per-country history (already stored)
Full offline dataset	API layer, build scripts	Data licence check
Translations	UI, content	String extraction

7. References

1. FlagForge source repository: https://github.com/KAUSHALCODER123/geo-guess
2. Next.js documentation: https://nextjs.org/docs
3. React documentation: https://react.dev
4. TypeScript documentation: https://www.typescriptlang.org/docs
5. Tailwind CSS documentation: https://tailwindcss.com/docs
6. REST Countries API: https://restcountries.com
7. Wikipedia REST API: https://en.wikipedia.org/api/rest_v1/
8. Open-Meteo: https://open-meteo.com/en/docs
9. TanStack Query: https://tanstack.com/query/latest
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11. Zod: https://zod.dev
12. Fuse.js: https://www.fusejs.io
13. Recharts: https://recharts.org
14. Radix UI primitives: https://www.radix-ui.com
15. Framer Motion: https://www.framer.com/motion
16. Vitest: https://vitest.dev
17. Playwright: https://playwright.dev
18. axe-core accessibility engine: https://github.com/dequelabs/axe-core
19. Web Content Accessibility Guidelines (WCAG) 2.2: https://www.w3.org/WAI/standards-guidelines/wcag/
20. Service Worker API (MDN): https://developer.mozilla.org/docs/Web/API/Service_Worker_API
21. Mulberry32 PRNG (public-domain algorithm by Tommy Ettinger), widely documented online.

A final bibliography should add access dates. Implementation claims in this report are limited to the repository source code.

8. Synopsis

Project title: FlagForge - Interactive World Atlas & Vexillology Quiz Arena.

Purpose: To provide unified country exploration, gamified geography practice and side-by-side comparison, available offline and without registration.

Users: Students, teachers and geography enthusiasts.

Platform: Browser-based responsive Progressive Web App (desktop and mobile).

Core technologies: Next.js 15, React 19, TypeScript, Tailwind CSS 4, TanStack Query, Zustand, Zod, Recharts, REST Countries, Wikipedia and Open-Meteo APIs.

Core modules:

- World Atlas Explorer with fuzzy search, filters and favorites
- Quiz Arena: six modes, three difficulty levels, scoring with streak multipliers
- Daily Global Challenge from a UTC-date seed
- Compare view with matrix and charts for up to three countries
- Stats dashboard: streaks, heatmap, region radar, weakest countries, achievements
- Country profiles with weather and Wikipedia summaries
- PWA: service worker, offline page, embedded fallback dataset

Outcome: An integrated, accessible and resilient geography learning application whose progress stays private on the learner's device.

Item	Value
Routes	8 page routes plus manifest, robots, sitemap and Open Graph image
Source code	About 16,400 lines of TypeScript/TSX including tests
Database	None (browser localStorage, 3 versioned stores)
Packages	23 runtime and 22 development dependencies (package.json)
Automated tests	4 Vitest files (58 tests) and 4 Playwright specs
Data sources	REST Countries, Wikipedia, Open-Meteo

Signature of the Student
Date: _____

Signature of the Guide
Place: _____